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Learning Check - Greedy Best First Search & Intro to A* Search

Instructions

Review Learning Objectives

coach

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Assignment details

Due
Oct 7, 11:59 PM IST

Attempts
3 left (5 attempts every 24 hours)

Try again

Submitted
Sep 29, 7:23 PM IST

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1. Which of the following search algorithms utilize a priority queue? Choose all that apply. **1 / 1 point**

- [x] A* Search

  Correct
  This option does utilize a priority queue.

- [ ] Breadth-First Search
- [x] Greedy Best-First Search

  Correct
  This option does utilize a priority queue.

- [x] Uniform Cost Search

  Correct
  This option does utilize a priority queue.

2. Which of the following is true about greedy best-first search? **1 / 1 point**

- ○ It always produces the most optimal solution when a solution is returned.
- ◉ At each step, the node with the minimal value as computed by the heuristic function is expanded.
- ○ It expands nodes farthest from the goal first to reach the solution quickly.
- ○ It is complete and always returns a solution.

  Correct
  This is within the definition of greedy best-first search.

3. Which of the following statements are true about A* search? Select all that apply. **1 / 1 point**

- [x] If for two admissible heuristics with $h_2(n) \geq h_1(n)$, then we can say h_2 dominates h_1.

  Correct
  A larger heuristic can better estimate the actual cost to reach the goal and is thus more accurate.

- [ ] The heuristic can have negative values.
- [x] Every consistent heuristic is also admissible.

  Correct
  Consistency implies admissibility.

- [ ] The graph-search of A* is optimal as long as the heuristic is admissible.